Enterprise Incident Response Simulation & Detection Platform

Track:

Infrastructure and Security - Cyber Security Incident Response Analyst

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1. Project Planning & Management

1.1 Project Proposal

The project titled "Enterprise Incident Response Simulation & Detection Platform" is designed to simulate real-world cyber-attacks in a controlled enterprise environment and evaluate the effectiveness of detection, response, and forensic investigation capabilities.

The system replicates a full Security Operations Center (SOC) environment, integrating the following capabilities:

- Attack simulation (Red Team activities)
- Detection engineering (Blue Team monitoring)
- Incident response workflows
- Digital forensics analysis
- Security visualization dashboard

The system follows the Incident Response Lifecycle:

- Preparation
- Detection & Analysis
- Containment
- Eradication
- Recovery
- Post-Incident Review

Project Objectives

- Simulate real-world enterprise cyber-attacks in a controlled environment
- Detect malicious activities and abnormal system behavior in real-time
- Implement structured incident response workflows
- Improve Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR)
- Map all attacks on the MITRE ATT&CK Framework
- Perform digital forensic investigations on simulated incidents
- Provide security dashboards for visualization and monitoring

Project Scope

- Enterprise network simulation (Windows Server, Clients, Kali Linux)
- Active Directory environment
- SIEM-based monitoring system
- Attack simulation scenarios
- Log collection and analysis
- Forensic investigation (memory and disk analysis)
- Dashboard visualization

1.2 Project Plan

Phase 1: Enterprise Lab Setup

Build a complete enterprise-like environment to simulate real-world infrastructure.

Components:

- Windows Server (Domain Controller – Active Directory)
- 2 Windows Client Machines (End Users)
- Kali Linux Machine (Attacker System)
- SIEM Platform: Wazuh or Splunk

Outcome:

- A fully functional enterprise network ready for attack simulation and monitoring.

Phase 2: Controlled Cyber Attack Simulation (Red Team Phase)

Simulate a realistic multi-stage cyber-attack inside the enterprise environment.

Attack Chain:

- Phishing Email Attack (Initial Access)
- Malware Execution (Payload Delivery)
- Privilege Escalation (User → Admin Access)
- Lateral Movement (Across network systems)
- Data Exfiltration (Sensitive data leakage)

Mapping:

- All attack stages are mapped to the MITRE ATT&CK Framework.

Outcome:

- Full documented attack lifecycle with logs and evidence.

Phase 3: Detection Engineering (Blue Team Defense)

Detect and analyze malicious activity in real-time using SIEM.

Activities:

- Writing detection rules (SIEM correlation rules)
- Generating real-time alerts
- Log analysis and correlation
- Mapping alerts to MITRE techniques
- Measuring security performance

KPIs:

- MTTD (Mean Time to Detect)
- MTTR (Mean Time to Respond)

Outcome:

- Operational detection system with alerts and security monitoring capability.

Phase 4: Digital Forensics Investigation

Investigate and reconstruct the full cyberattack after detection.

Analysis Sources:

- System Logs
- Memory Dumps
- Disk Artifacts

Tools:

- Autopsy (Disk Forensics)
- Volatility (Memory Analysis)

Outputs:

- Full attack timeline reconstruction
- Indicators of Compromise (IOCs)
- Root Cause Analysis

Outcome:

- Complete forensic report explaining how the attack happened from start to end.

Phase 5: SOC Dashboard & Visualization

Build a Security Operations Center (SOC) dashboard to visualize all incidents.

Dashboard Includes:

- Number of security incidents
- Types of attacks
- Attack phases (progression view)
- Detection time (MTTD)
- Response time (MTTR)

Tools:

- Elastic Stack
- Power BI (alternative)
- SIEM built-in dashboards (Wazuh/Splunk)

Outcome:

- Interactive dashboard representing full enterprise security visibility.

1.3 Task Assignment & Roles

Roles and responsibilities are distributed across the team to ensure coverage of all project phases, including lab setup, attack simulation, detection engineering, forensic analysis, and dashboard development. Detailed role assignments are tracked in the project management log.

1.4 Risk Assessment & Mitigation Plan

Risk	Mitigation Strategy
SIEM overload from logs	Filtering rules and log optimization
False positive alerts	Fine-tuned detection thresholds
Attack simulation instability	Isolated virtual lab environment
Data loss during analysis	VM snapshots and backups
Integration issues	Continuous testing and validation

1.5 Key Performance Indicators (KPIs)

- Mean Time To Detect (MTTD)
- Mean Time To Respond (MTTR)
- False Positive Rate
- Detection Coverage (% of MITRE ATT&CK techniques detected)
- Incident Resolution Time
- Alert Accuracy Rate

2. Literature Review

2.1 Feedback & Evaluation

Modern cybersecurity incident response systems are evaluated based on the following criteria:

- Detection accuracy
- Response efficiency
- Log correlation capability
- System reliability
- Analyst usability

Enterprise SOC environments rely heavily on centralized monitoring systems such as Wazuh and Splunk. These tools enable real-time monitoring, threat detection, and incident correlation.

2.2 Suggested Improvements

Existing systems can be improved through the following initiatives:

- Enhanced simulation-based training environments
- Better visualization of attack chains
- Stronger integration with MITRE ATT&CK mapping
- Improved forensic automation capabilities
- Reduction of false positive alerts
- More interactive SOC dashboards

2.3 Final Grading Criteria

Academic evaluation is based on the following dimensions:

- Quality of system design and documentation
- Completeness of implementation
- Depth of cybersecurity understanding
- Accuracy of testing and validation
- Quality of presentation and demonstration

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholders are the individuals or groups who are directly or indirectly affected by the system. In this project, the main stakeholders include:

- **monitor and investigate security alerts** SOC Analysts:
- **manage infrastructure and logs** System Administrators:
- **learn through simulation** Cybersecurity Students/Trainers:
- **handle security incidents** Incident Responders:
- **configure detection rules and SIEM** Security Engineers:

3.2 User Stories & Use Cases

- As a SOC analyst, I want to detect suspicious activities early to respond quickly.
- As a system admin, I want to monitor logs to ensure system stability.
- As a student, I want to simulate attacks to understand real-world incidents.
- As a forensic analyst, I want to reconstruct attack timelines.
- As a security engineer, I want to map alerts to MITRE ATT&CK techniques.

3.3 Functional Requirements

- The system shall simulate cyber attacks in a controlled environment.
- The system shall detect malicious activities in real-time.
- The system shall log all security events centrally.
- The system shall generate alerts for suspicious behavior.
- The system shall map detected threats to MITRE ATT&CK techniques.
- The system shall support forensic investigation of incidents.
- The system shall visualize security events in dashboards.

3.4 Non-Functional Requirements

- Performance: near real-time detection and response
- Security: secure handling of logs and forensic data
- Usability: simple interface for analysts and students
- Reliability: continuous operation during simulations
- Scalability: support multiple concurrent attack scenarios
- Auditability: all actions must be fully logged

4. System Analysis & Design

4.1 Problem Statement & Objectives

Problem Statement

Modern organizations face increasing cyber threats that require rapid detection, analysis, and response. However, many environments lack a structured simulation platform to train cybersecurity teams on real-world attack scenarios and incident response procedures. This project addresses this gap by simulating a complete enterprise environment where cyberattacks are executed, detected, analyzed, and investigated using professional cybersecurity tools.

Objectives

- Simulate real-world enterprise cyberattacks in a controlled environment
- Detect malicious activities using SIEM solutions
- Apply structured incident response methodology
- Perform forensic investigation on security incidents
- Map all attack stages to the MITRE ATT&CK Framework
- Visualize security events through dashboards

4.2 Use Case Diagram & Descriptions

Actors

- SOC Analyst
- System Administrator
- Attacker (Kali Linux)
- Security Engineer

Main Use Cases

- Execute cyberattack simulation
- Monitor system logs
- Detect suspicious activity
- Generate security alerts
- Investigate incidents
- Perform forensic analysis
- View security dashboard

4.3 Functional & Non-Functional Requirements

Functional Requirements

- The system shall simulate enterprise-level cyberattacks.
- The system shall collect and analyze logs from all endpoints.
- The system shall generate real-time security alerts.
- The system shall map detected incidents to MITRE ATT&CK techniques.

- The system shall support forensic investigation processes.
- The system shall provide visualization dashboards.

Non-Functional Requirements

- Performance: near real-time detection and response
- Security: secure storage and handling of logs and forensic data
- Reliability: continuous monitoring during simulations
- Usability: simple interface for SOC analysts and students
- Scalability: ability to handle multiple attack scenarios

4.4 Software Architecture

The system follows a centralized security monitoring architecture organized across the following layers:

- Endpoint Layer (Windows Clients & Server)
- Attack Layer (Kali Linux)
- Collection Layer (Log forwarding agents)
- SIEM Layer (Wazuh / Splunk)
- Analysis Layer (Detection rules & correlation engine)
- Visualization Layer (Dashboard)

Architecture style: Centralized Security Monitoring (SOC-based architecture).

4.5 Database Design & Data Modeling

ER Diagram – Conceptual Entities

Entity	Attributes (PK/FK)	Relationships
Users	ID (PK), Name, Role	Monitor Systems
Systems	ID (PK), Type, IP Address	Generate Logs
Logs	ID (PK), System_ID (FK), Time, Event	Trigger Alerts
Alerts	ID (PK), Log_ID (FK), Severity, Desc.	Form Incidents
Incidents	ID (PK), Alert_ID (FK), Status, Time	Linked to Attack Events

4.6 Data Flow & System Behavior

Context Level
Users ↔ System (Monitor & Analyze).

Level 1 & Sequence Diagram

Attack → Logs Generated → SIEM Analysis → Alert Triggered → Analyst Investigation → Forensic Analysis.

Activity Diagram

Start → Simulation → Logging → Monitoring → Detection → Alert → Investigation → End.

State Diagram

[Detected] → [Under Investigation] → [Contained] → [Resolved].

Class Architecture
Users manage Incidents; Systems generate Logs; Logs trigger Alerts; Alerts create Incidents.

4.7 UI/UX Design & Prototyping

The system includes a dashboard-based UI providing the following views:

- Security alerts overview
- Incident timeline
- Attack visualization
- System status

UI/UX Guidelines

- Simple and minimal SOC-style interface
- Color coding for severity levels: Low, Medium, High, Critical
- Clear visualization of attack stages
- Responsive dashboard design
- Focus on usability for SOC analysts

4.8 System Deployment & Integration

Technology Stack

- Windows Server (Active Directory)
- Windows Clients
- Kali Linux
- SIEM: Wazuh or Splunk
- Forensics: Autopsy, Volatility
- Visualization: ELK Stack / Power BI

Deployment Diagram

- Internal Enterprise Network (Isolated Lab Environment)
- SIEM Server connected to all endpoints
- Attacker machine isolated in a separate network segment
- Centralized log collection system

Component Diagram

- Endpoint Monitoring Agents
- SIEM Engine
- Detection Rules Engine
- Alert Management Module
- Dashboard Visualization Module

4.9 Additional Deliverables

API integration deliverables are not applicable unless the system is extended to include external API endpoints.

Testing & Validation

- Unit testing for detection rules
- Integration testing between endpoints and SIEM
- Simulation testing for attack scenarios
- Validation of alert accuracy and response

Deployment Strategy

- Virtualized environment setup using VM tools
- Isolated lab network for safe attack simulation
- Central SIEM deployment
- Continuous log monitoring pipeline